

Semester	Course Code	Course Category	Hours/ Week	Credits	Marks for Evaluation		
					CIA	ESE	Total
I	23MCA1CC1	CORE – I	4	3	25	75	100
Course Title		Programming In Java					

SYLLABUS		
Unit	Contents	Hours
I	Java Buzzwords – Class Fundamentals – Declaring Objects – Introducing Methods – Constructors – This keyword – Garbage Collection – Overloading Methods – Argument Passing – Recursion – Access Control Understanding static – final. Inheritance Basics – Using super – Method overriding – Dynamic Method Dispatch – Using Abstract Classes – * Final with Inheritance * – Object class.	12
II	Defining a Package– Access Protection – Importing Packages – Interfaces. Exception Handling Fundamentals – Exception Types – Using try and catch– Multiple catch Clauses – Nested try Statements– throw – throws – finally – Java’s Built-in Exceptions – * Creating Own Exception Subclasses *. The Java Thread Model – Creating a Thread – Creating Multiple Threads-Thread Priorities– Synchronization – Inter-thread Communication.	12
III	String Handling – The lang package: Primitive Type Wrappers . The Collections Framework: List, Set – ArrayList – LinkedList – Vector – Stack– HashTable – StringBuilder – Date – Calendar - Random – Scanner. The I/O Classes and Interfaces - File – The Byte Streams: InputStream – OutputStream – *FileInputStream – File Output Stream* – Print Stream– DataOutputStream – DataInputStream –The Character Streams: Reader– Writer – FileReader – FileWriter – PrintWriter – Serialization.	12
IV	Networking Basics – InetAddress –TCP/IP Client Sockets – URL – URLConnection – *TCP/IP Server Sockets* – Datagrams – Java Database Connectivity: Establishing a connection – Creation of Data Tables – Entering Data into Tables – Table Updating – Use of PreparedStatement – Obtaining Metadata.	12
V	Event Handling: Delegation Event Model – Event Classes – Event Listener Interfaces – Layout Managers - Components and Containers – The Swing Packages – Exploring Swing: JLabel and ImageIcon – *JTextField – JButton – Check boxes* - Radio Buttons – JTabbedPane– JScrollPane – JList – JComboBox – Jtrees - JTable.	12
VI	Current Trends (For CIA only) – Contemporary developments related to the course during the semester concerned.	

..... Self Study

Text Book(s):
1. Herbert Schildt, Java The Complete Reference, Eleventh Edition, McGraw-Hill Education,2019. 2. C. Muthu, Programming with Java, Vijay Nicole imprints private Limited, 2004(for JDBC only).
Reference Book(s):
1. Sachin Malhotra and Saurabh Chaudhary, Programming in Java, Oxford University Press, 2018. 2. Daniel Liang, Introduction to Java Programming, Tenth Edition, Pearson, 2015.
Web Resource(s):
1. https://www.javatpoint.com/java-tutorial 2. https://in.coursera.org/specializations/java-programming

Course Outcomes		
Upon successful completion of this course, the student will be able to:		
CO No.	CO Statement	Cognitive Level (K-Level)
CO1	Remembering the fundamental concept of OOPs in Java language	K1
CO2	Illustrate the concepts like packages, interfaces, exception handling, multithreading, Collection, I/O and Networking classes and database connectivity.	K2
CO3	Apply appropriate problem solving strategies	K3
CO4	Design GUI based applications.	K5
CO5	Create the Java applications to solve the real time problems.	K6

Relationship Matrix:

Course Outcomes (COs)	Programme Outcomes (POs)					Programme Specific Outcomes (PSOs)					Mean Score of COs
	PO1	PO2	PO3	PO4	PO5	PSO1	PSO2	PSO3	PSO4	PSO5	
CO1	3	2	3	2	2	2	3	2	3	2	2.4
CO2	2	0	3	3	0	3	2	0	3	3	1.9
CO3	2	3	3	0	3	2	0	3	2	3	2.1
CO4	3	3	3	3	3	3	2	3	2	2	2.7
CO5	2	0	3	2	2	3	3	3	3	2	2.3
Mean Overall Score											2.28
Correlation											Medium

Mean Overall Score = Sum of Mean Score of COs / Total Number of Cos

Mean Overall Score	Correlation
< 1.5	Low
≥ 1.5 and < 2.5	Medium
≥ 2.5	High

Course Coordinator: Dr. M. Mohamed Surputheen